

A. PDDL Task Specification

Domain: example action schema

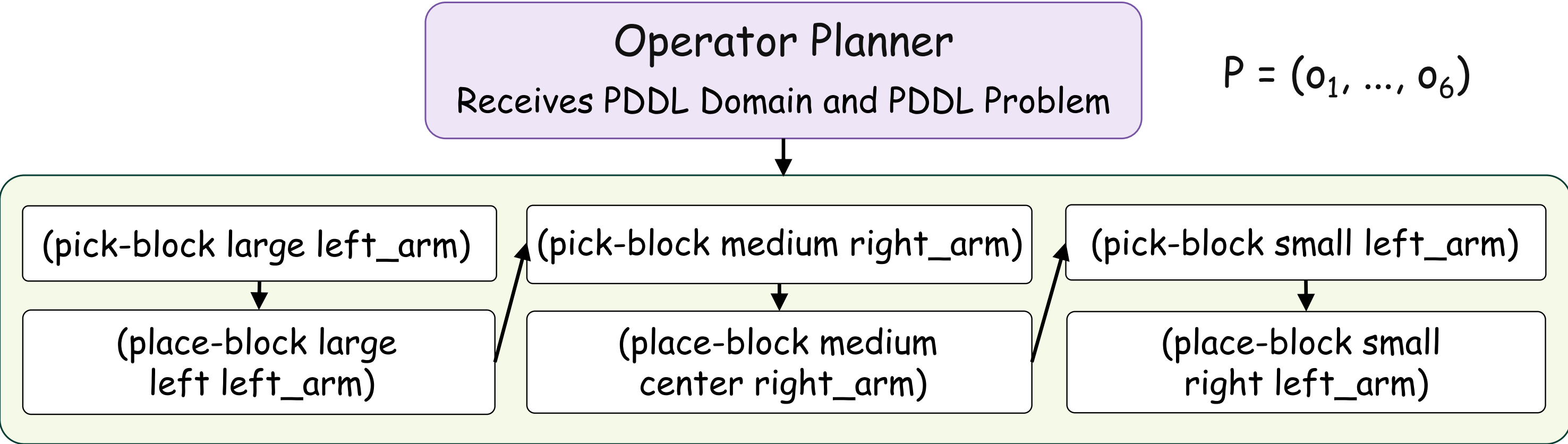
```
(:action place-block
:parameters (?b - block ?p - position ?arm - arm)
:precondition
  (and
    (holding ?arm ?b)
    (clear ?p))
:effect
  (and
    (at ?b ?p)
    (handempty ?arm)
    (not (holding ?arm ?b))
    (not (clear ?p))))
```

Problem: task instance

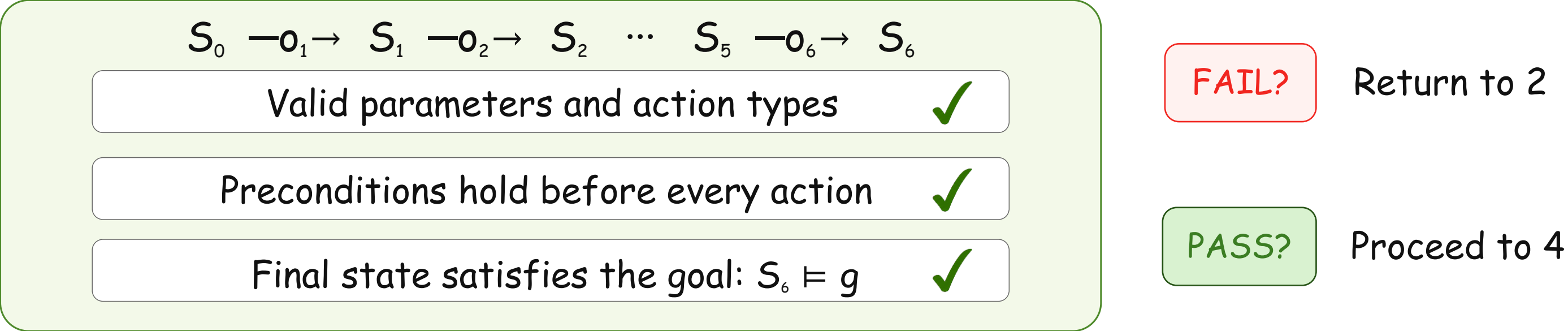
```
(define (problem rank-blocks)
  (:domain block-placement)
  (:objects
    large medium small - block
    left center right - position
    left_arm right_arm - arm)
  (:init
    (handempty left_arm) (handempty right_arm)
    (clear left) (clear center) (clear right))
  (:goal
    (and (at large left)
          (at medium center)
          (at small right))))
```

Domain defines reusable actions; Problem provides S_0 and g

B. Operator Planner → Linear Reference Plan



C. VAL Plan Checking



D. BuildGraph → Initial Task DAG G

